

Total No. of Questions : 10]

SEAT No. :

P3752

[5561]-151

[Total No. of Pages : 3

B.E. (Mechanical)

**DESIGN OF PUMPS, BLOWERS AND COMPRESSORS
(2012 Pattern) (Semester - II) (Elective - IV) (End Semester) (402050 C)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Use of calculator, steam table is allowed.
- 5) Assume suitable data if necessary.

- Q1) a)** Explain dimensionless parameters used in compressible flow machines. [6]
b) Explain performance characteristics of fan, blower and compressors. [4]

OR

- Q2) a)** What is specific speed? Explain its significance? [4]
b) A single acting reciprocating pump running at 50 rpm delivers $0.01 \text{ m}^3/\text{s}$ of water. The diameter of the piston is 200 mm and stroke length 400 mm determine [6]
i) The theoretical discharge of the pump
ii) Co-efficient of discharge and
iii) Slip and the percentage slip of the pump.

- Q3) a)** Explain indicator diagram in reciprocating pumps. [4]
b) A single acting reciprocating pump has a plunger diameter 250 mm and stroke of 450 mm and it is driven with S.H.M. at 60 rpm. The length and diameter of delivery pipe are 60 m and 100 mm respectively. Determine the power saved in overcoming friction in the delivery pipe by fitting an air vessel on the delivery side of the pump. Assume friction factor = 0.01. [6]

OR

- Q4) a)** Explain various forms of corrosion occurring in hydraulic machines. [5]
b) Explain general design consideration of Pump impeller. [5]

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Q5) a) What are main causes for noise generation? What are methods for reducing the fan noise? [8]

b) Explain the different losses in fans, blowers and compressors. [8]

OR

Q6) a) Explain the different fan stage parameters with usual notations. [8]

b) Explain in detail the design procedure, selection and optimization of blower. [8]

Q7) a) What is surging and stalling? What are its effect? How it is developed? [8]

b) A centrifugal blower with a radial impeller produces a pressure equivalent to 100 cm column of water. The pressure and temperature at its entry are 0.98 bar and 310 K. The electric motor driving the blower runs at 3000 rpm. The efficiencies of the fan and drive are 82 % and 88 % respectively. The radial velocity remains constant and has a value of $0.2u_2$. The velocity at the inlet eye as $0.4u_2$. If the blower handles 200 m³/min of air at the entry condition determine :- [10]

i) Power required by the electric motor

ii) Impeller diameter

iii) Inner diameter of the blade ring

iv) Air angle at entry

v) Impeller widths at entry and exit

vi) Number of impeller blades

vii) The Specific speed.

OR

Q8) a) Explain function of an aerofoil and discuss characteristics curve of airfoil. [8]

b) An axial fan stage consisting of only a rotor has the following data :-

Rotor blade air angle at exit: 10°

Tip diameter: 60 cm

Hub diameter: 30 cm

Rotational speed: 960 rpm

Power required: 1 kW

Flow coefficient: 0.245

(Inlet flow conditions $P_1 = 1.02$ bar and $T_1 = 316$ K). Determine the rotor blade angle at the entry, the flow rate, stage pressure rise, overall efficiency, degree of reaction, and specific speed. [10]

- Q9) a)** Explain Enthalpy-Entropy diagram of a compressor in detail. [8]
- b)** Explain the following terms [8]
- i) degree of reaction
 - ii) slip factor
 - iii) choking.

OR

- Q10)a)** Explain the performance characteristics curves of Centrifugal Compressor. [8]
- b)** What is work done factor for an axial flow compressor stage? How does it vary with no of stages? [8]

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